

Experimental study of the impact of virtual reality on fiber bundle cleaning and annotation

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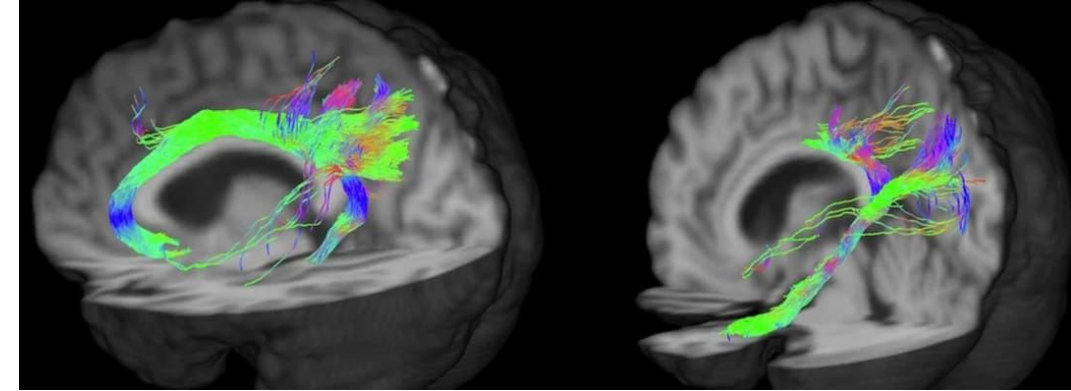
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Why this project?

- dMRI tractography reconstructs 3D white-matter pathways
- Automated bundle extraction remains imperfect and may include erroneous or missing fibers
- Manual curation of fiber bundles is often required to improve anatomical accuracy
- On a desktop display : this task is time-consuming and spatially demanding



Why Virtual Reality (VR) ? :

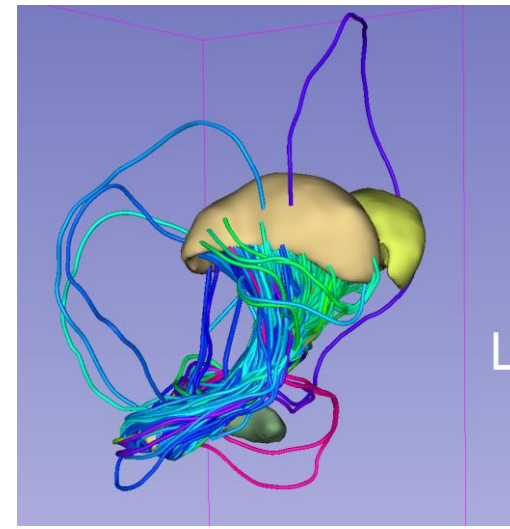
- Stereoscopic 3D view
- Better depth perception
- Direct interaction with objects

Objective :

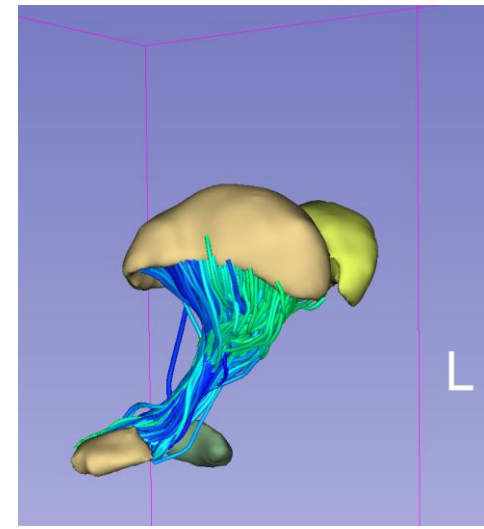
To compare VR and desktop for manual cleaning of fiber images in order to assess task efficiency, cleaning accuracy, reproducibility, and user satisfaction

Methods : data preparation

- 5 Datasets from Human Connectome Project – Young Adult
- 4 tractoms generated for each HCP-YA data :
tract to be cleaned and cleaned reference tract
- 6 trials per condition (VR/Desktop) with 2 repeated tractoms



Tract to be cleaned



Cleaned reference tract

Materials :

- 3D Slicer, SlicerVR, Slicer dMRI
- Meta Oculus Quest 3
- Laptop : ASUS ROG Zephyrus

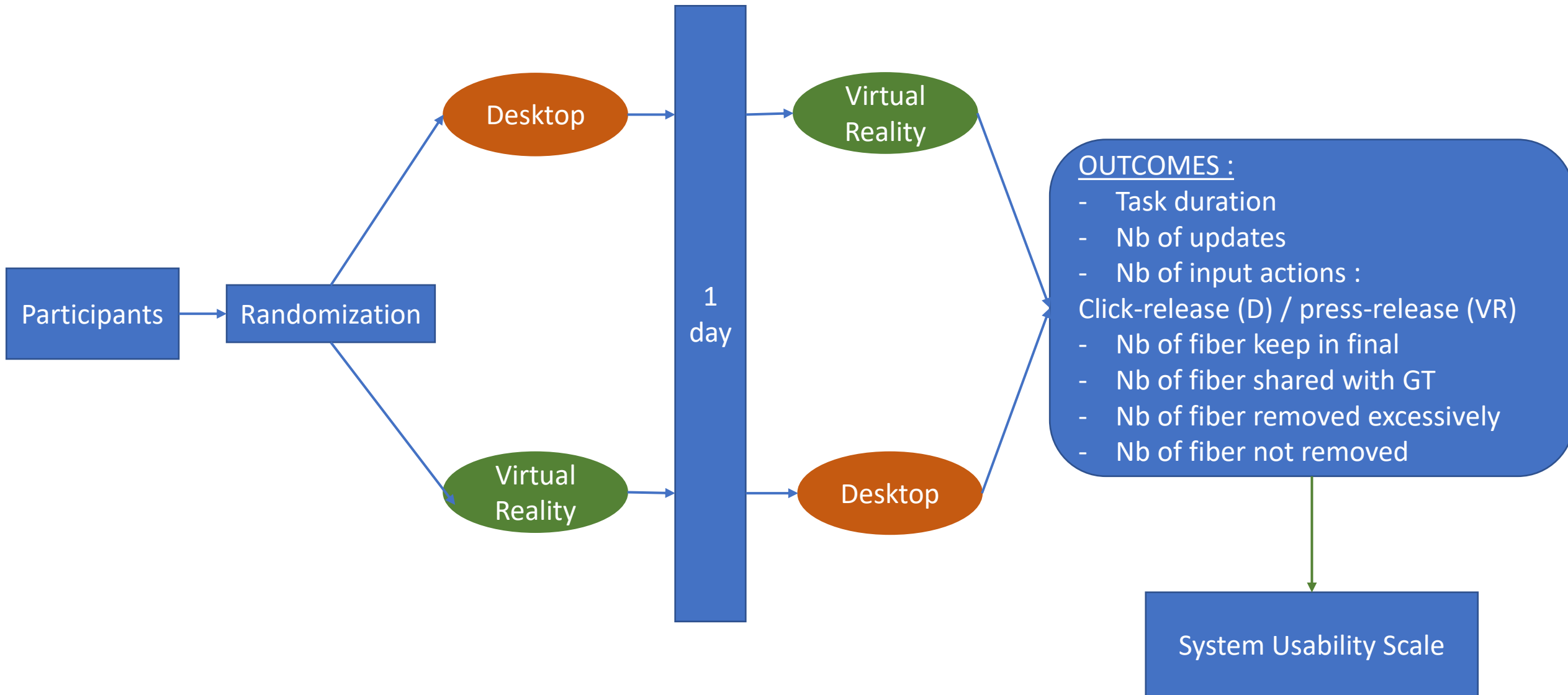
(AMD Ryzen 9 processor, an NVIDIA GeForce RTX 4060 Laptop GPU, and 16 GB RAM)

User interface :

- Desktop : python module using Slicer dMRI functionalities (**ROI**)
- VR : python module using a **cube model**

Same task on both condition (VR/Desktop)

Methods : Counterbalanced crossover design



Methods : data analysis

- Adaption of the Dice coefficient to tractograms : global agreement between participant's cleaned tractogram and the expert reference

$$Dice = \frac{2|G \cap M|}{|G| + |M|}$$

G : set of ground truth fibers

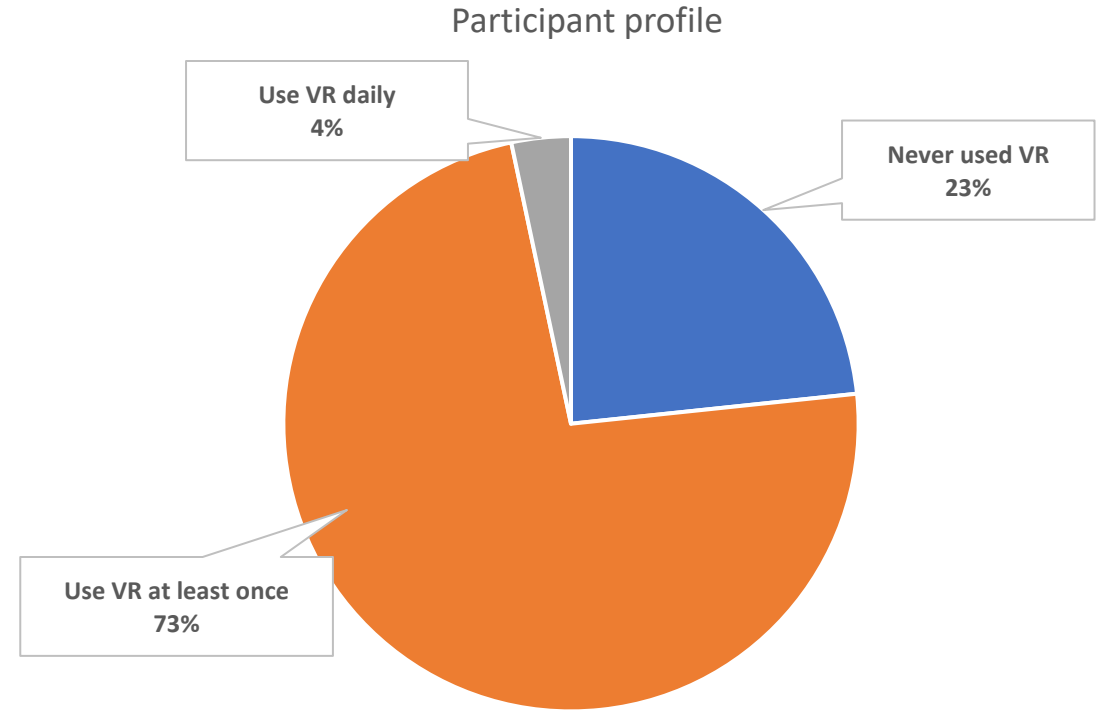
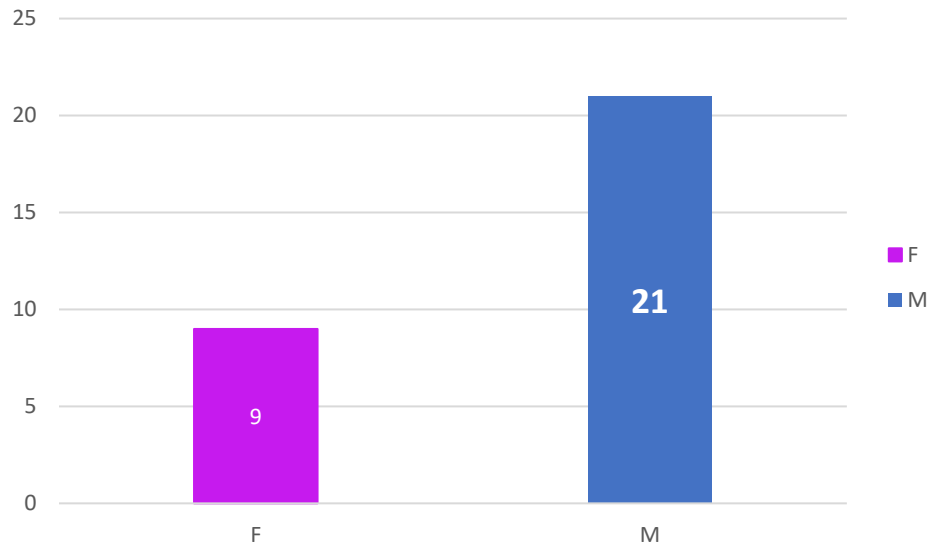
M : set of fibers retained by participants

- Linear mixed models (LMM) :
effect of visualization mode (Desktop / VR) on Dice and log transform task duration
- Interclass correlation coefficient (ICC) :
intra-participant repeatability

Results : participant profile

30 participants recruited

(mean age: 28.4 ± 7.4 years; 9 females, 21 males)

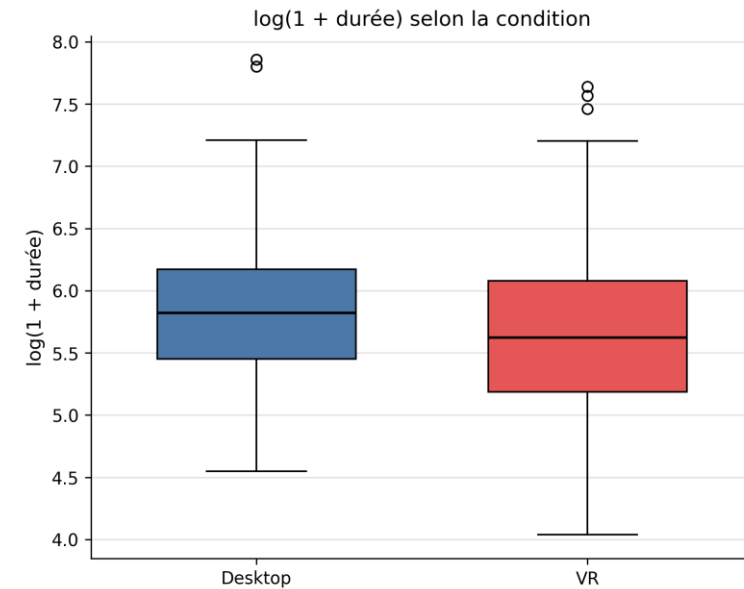
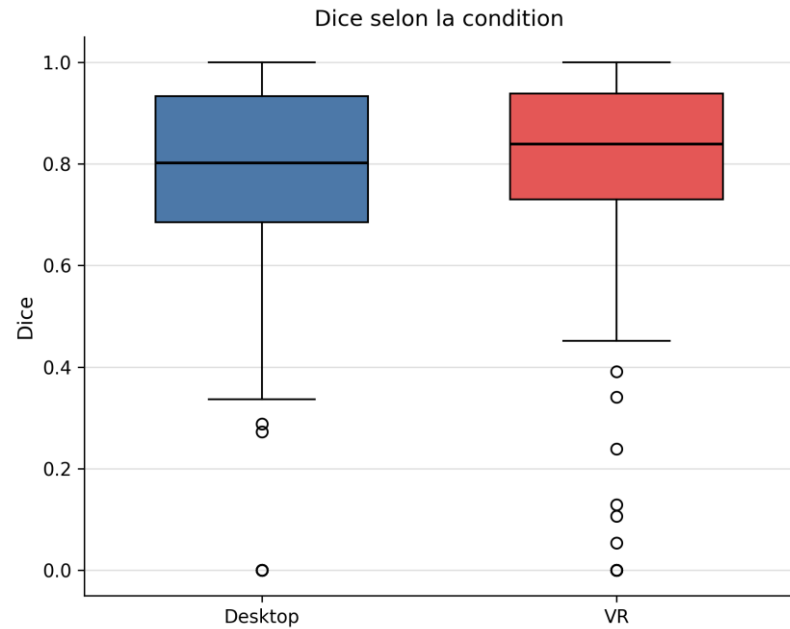


Descriptive results

Table 1 Descriptive statistics of performance metrics in the Desktop and VR conditions

Variables	Desktop Mean $\pm$ SD	VR Mean $\pm$ SD	Desktop Median [Q1–Q3]	VR Median [Q1–Q3]
Dice score	0.75 $\pm$ 0.12	0.78 $\pm$ 0.10	0.78 [0.69–0.86]	0.81 [0.69–0.87]
Task duration (s)	415.03 $\pm$ 173.72	377.13 $\pm$ 170.62	376.51 [287.70–518.44]	364.75 [227.57–471.50]
Number of interactions	44.92 $\pm$ 16.68	43.96 $\pm$ 19.50	42.00 [33.79–52.96]	40.25 [31.17–49.58]
Number of updates	13.84 $\pm$ 4.76	26.67 $\pm$ 14.19	12.96 [11.17–16.62]	22.92 [19.71–29.29]

Results : Effects of visualization on Dice score and task duration

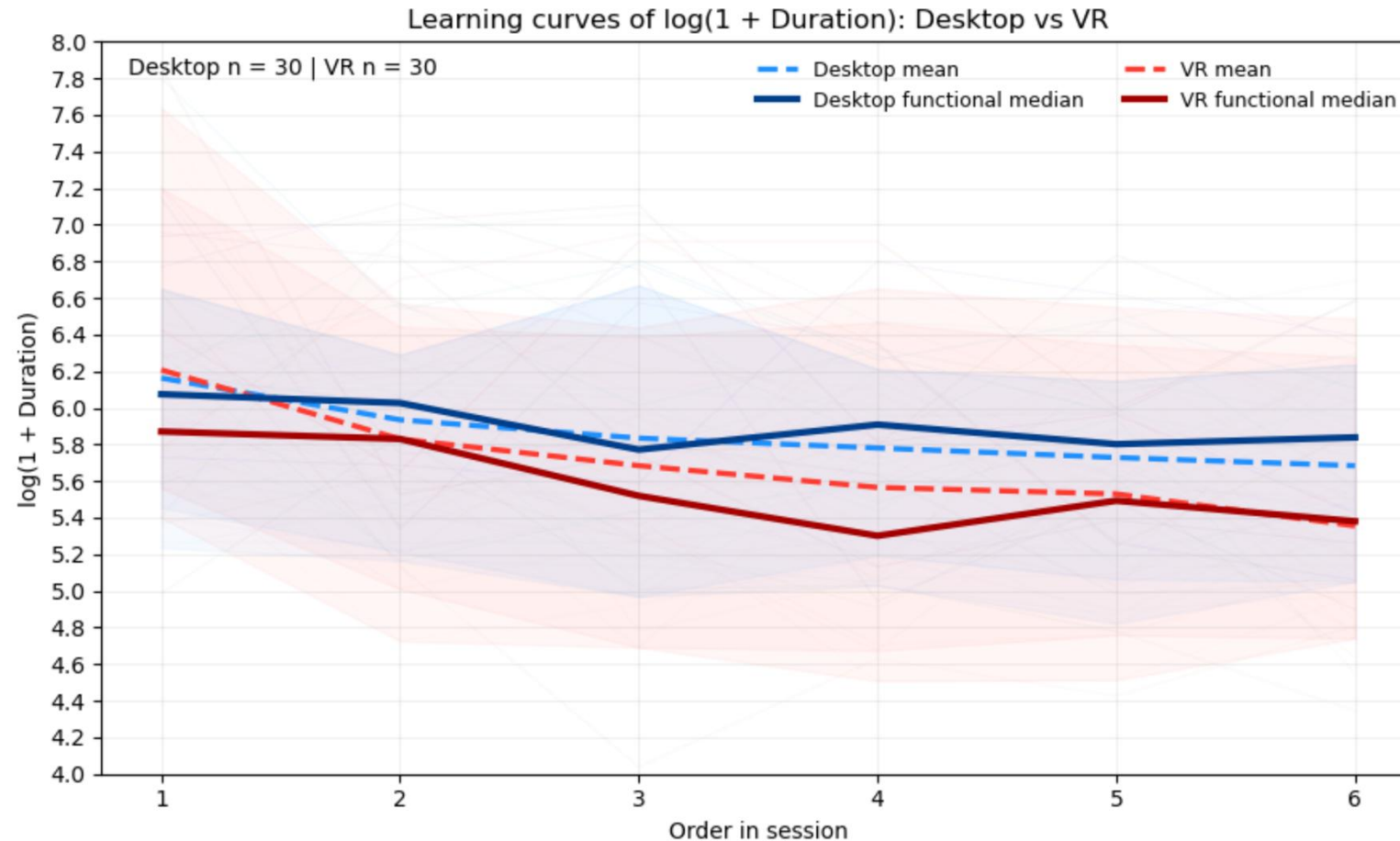


Outcome	Effect	P value
Dice	$\beta = 0,038$	0,389
Time (log)	$\beta = - 0,192$	0,032
Interaction condition $\times$ case	Non significant	$> 0,05$

17.5% reduction in task duration in VR relative to Desktop

Results : learning curve

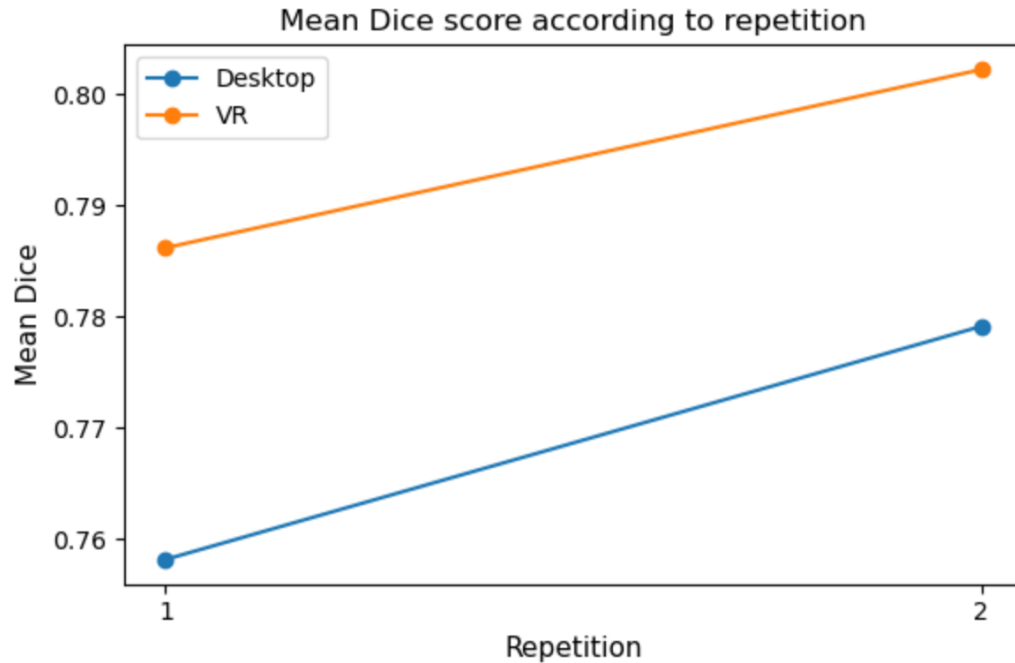
Mesure	effect	p	Interpretation
Order in session	$\beta = -0,125$	$<0,001$	Learning effect



Results : post hoc analyses

Outcome / cases	n	Mean VR	Mean Desktop	Diff (VR – Desktop)	p-value (Holm)
<i>Dice score</i>					
Case 1	30	0.805	0.764	+ 0.041	0.027
Case 2	30	0.766	0.772	- 0.005	0.971
Case 3	28	0.763	0.639	+ 0.033	0.141
Case 4	30	0.855	0.834	+ 0.021	0.620
<i>Log-duration</i>					
Case 1	30	5.883	6.007	- 0.124	0.288
Case 2	30	5.245	5.562	- 0.317	0.010
Case 3	28	5.973	6.060	- 0.087	0.293
Case 4	30	5.605	5.771	- 0.166	0.288

Results : effect of the repetition on Dice



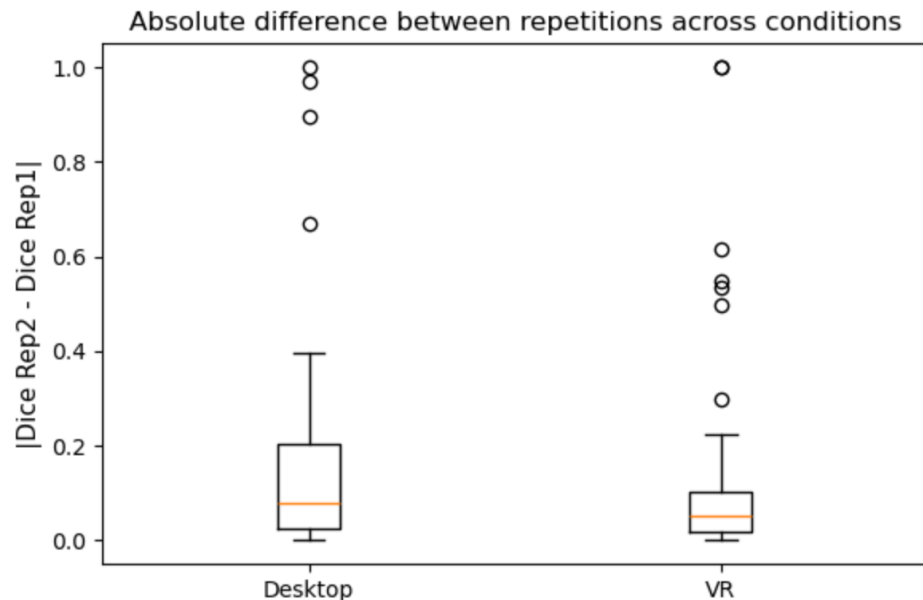
Condition	Dice Repetition1	Dice Repetition2
Mean (± SD) Desktop	0,758 ± 0,209	0,779 ± 0,213
Mean (± SD) VR	0,786 ± 0,225	0,802 ± 0,220

Effect	β	P-value
Condition (VR vs Desktop)	+0,028	0,425
Repetition 2 vs 1	+0,021	0,555
Condition × repetition	-0,005	0,922

→ The Dice score increased slightly on the second repetition in both conditions, but there was no significant effect of repetition, nor any interaction with the condition

Results : reproducibility intra-participant

Condition	ICC(2,1)	ICC(3,1)	Mean Dice Rep2 - Dice Rep1	Median Dice Rep2 - Dice Rep1
Desktop	0,106	0,104	0,166	0,081
VR	0,385	0,382	0,122	0,051



→ Reproducibility across repetitions was low in both conditions, with a higher ICC observed in VR than in Desktop

→ The absolute difference between repetitions tended to be **smaller in VR**, suggesting better within-participant consistency, although the difference was not statistically significant

Results : System Usability Scale

Participant	SUS Desktop	SUS VR
1	72,50	77,50
2	70,00	67,50
3	75,00	90,00
4	47,50	87,50
5	65,00	NA
6	75,00	85,00
7	65,00	90,00
8	82,50	90,00
9	62,50	82,50
10	25,00	72,50
11	50,00	82,50
12	42,50	77,50
13	70,00	67,50
14	35,00	80,00
15	70,00	70,00
16	40,00	70,00
17	55,00	65,00
18	97,50	75,00
19	70,00	80,00
20	27,50	70,00
21	45,00	80,00
22	67,50	90,00
23	72,50	70,00
24	65,00	75,00
25	55,00	65,00
26	50,00	75,00
27	27,50	92,50
28	57,50	75,00
29	30,00	65,00
30	67,50	87,50

	n	Desktop (M±SD)	VR (M±SD)	CI 95%	t(df)	P-value
SUS	29	60.00 ± 19.45	77.50 ± 10.26	[9.38 ; 25.62]	4.42 (28)	0.000136

- VR had a **higher SUS** score than Desktop, with a mean difference of **+17.5 points**
- This difference was **statistically significant**

Conclusion

- VR has no clear significant advantage for Dice accuracy
- Cleaning time was reduced by 17% in VR
- Reproducibility remained low overall, but tended to be better in VR
- User satisfaction in terms of usability was significantly higher in VR

VR was better accepted by users and showed encouraging consistency trends, even though no clear accuracy advantage was demonstrated

Futur work :

- Confirm these findings in a larger cohort
- Evaluate additional fiber bundles and tractogram configurations
- Optimize VR interaction for tractography cleaning
- Assess learning effects and reproducibility over longer use

THANK YOU!